

PROJECT DEVELOPMENT PHASE – PART 2

Dashboard and Story Development

Dashboard Development

After completing the data preprocessing and creating individual visualizations, three interactive dashboards were developed in Tableau Desktop. These dashboards provide a comprehensive analysis of electricity consumption patterns across different states, regions, quarters, months, and years. Interactive filters were implemented to enable dynamic data exploration and support effective decision-making.

Dashboard 1: State and Regional Electricity Consumption Analysis

Description

This dashboard provides a comprehensive overview of electricity consumption across different states and regions for the years **2019** and **2020**. It includes state-wise filled maps, regional electricity distribution, state-wise consumption comparison, and electricity usage before and during the COVID-19 lockdown.

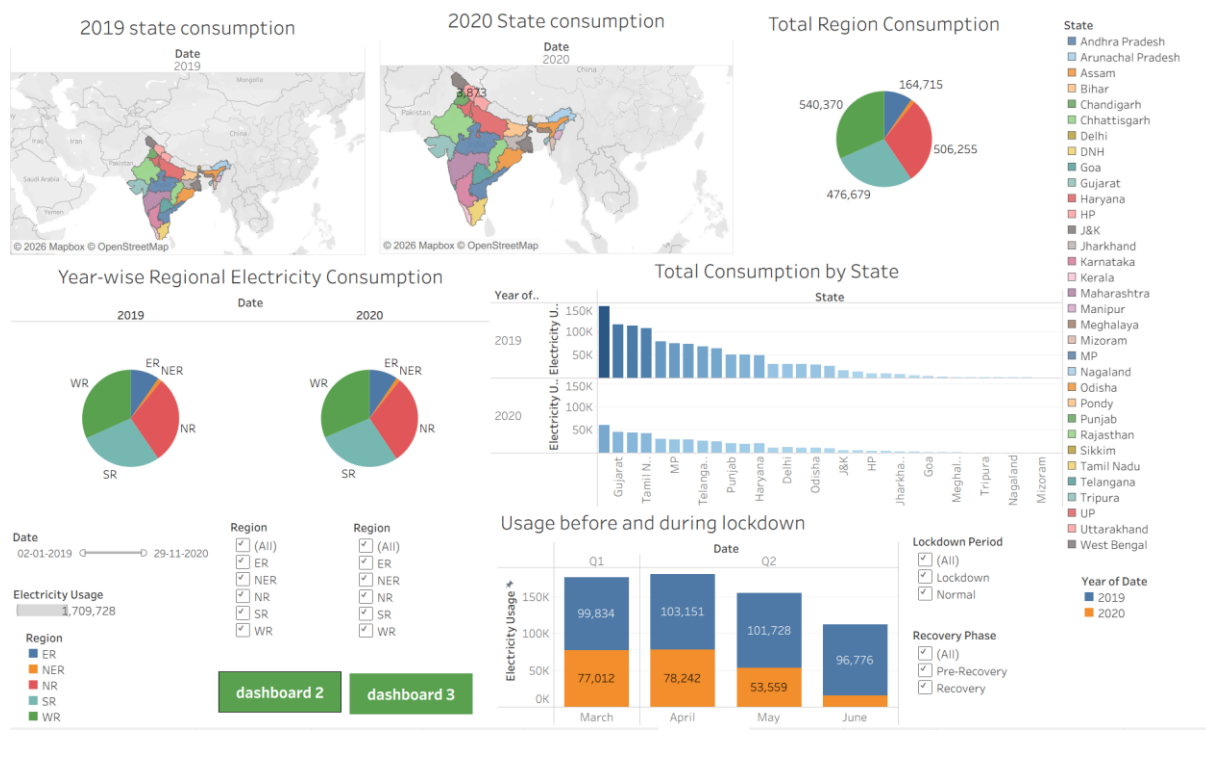
Visualizations Included

- State-wise Electricity Consumption (2019 Filled Map)
- State-wise Electricity Consumption (2020 Filled Map)
- Total Region Consumption (Pie Chart)
- Year-wise Regional Electricity Consumption
- Total Electricity Consumption by State
- Electricity Usage Before and During Lockdown

Filters Used

- Date
- Region
- State
- Lockdown Period
- Recovery Phase

Dashboard Screenshot



Dashboard 2: Region-wise and Top/Bottom State Analysis

Description

This dashboard focuses on regional electricity consumption and identifies the highest and lowest electricity-consuming states. It also presents yearly electricity usage trends to facilitate comparative analysis across regions.

Visualizations Included

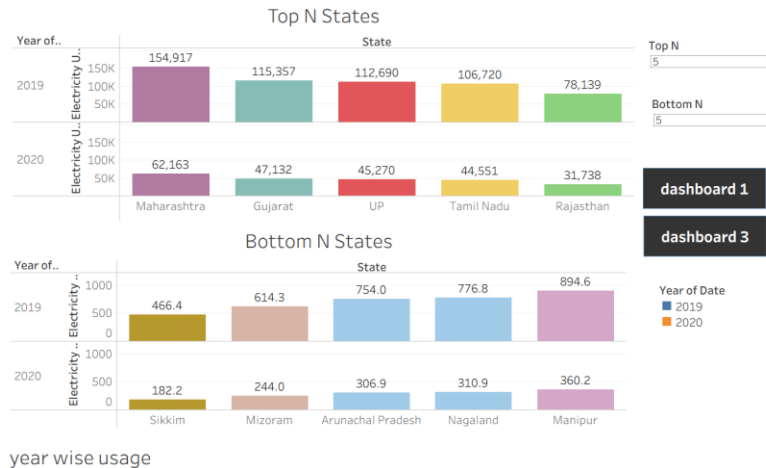
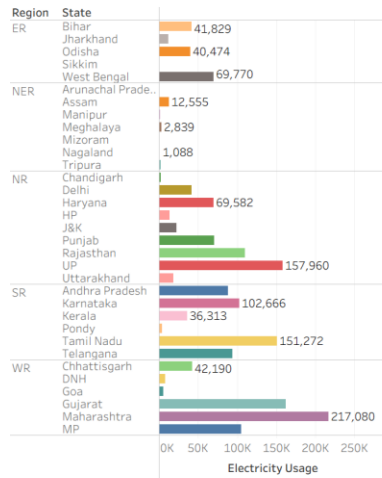
- Region-wise State Electricity Consumption
- Top 5 Electricity Consuming States
- Bottom 5 Electricity Consuming States
- Year-wise Electricity Usage Trend

Filters Used

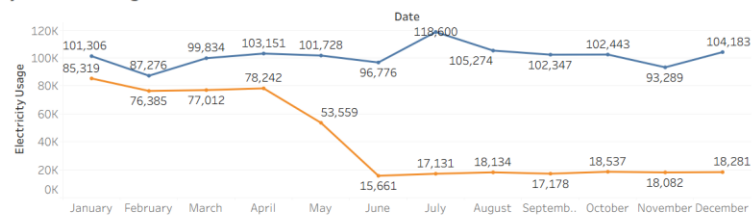
- Top N
- Bottom N
- Year

Dashboard Screenshot

regionwise state consumption



year wise usage



Dashboard 3: Quarter-wise and Month-wise Electricity Consumption Analysis

Description

This dashboard presents electricity consumption trends across different quarters and months for the years **2019** and **2020**. It helps users compare seasonal electricity consumption patterns and analyze monthly electricity usage.

Visualizations Included

- Quarter-wise Electricity Consumption
- Month-wise Electricity Consumption (2019)
- Month-wise Electricity Consumption (2020)
- Month-wise Electricity Usage Trend

Filters Used

- State
- Month
- Electricity Usage

Dashboard Screenshot

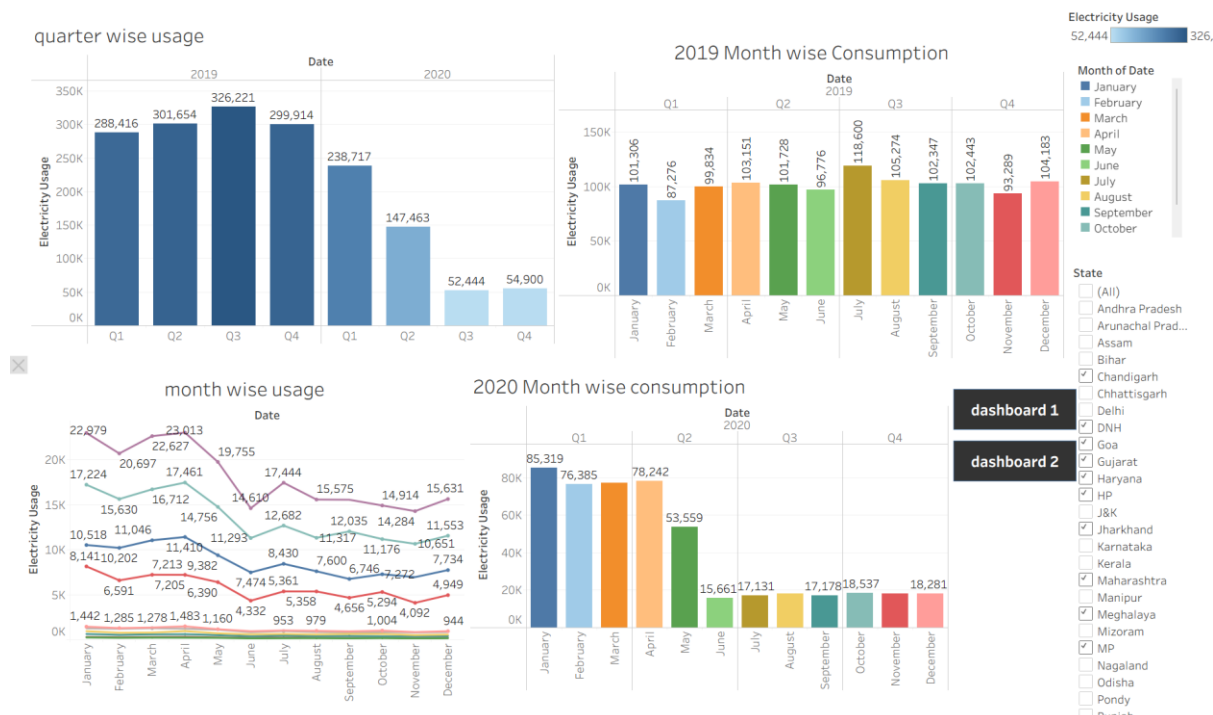


Tableau Story

Description

A Tableau Story was created by organizing the three interactive dashboards into a logical sequence. The story enables users to explore electricity consumption patterns step by step, beginning with state and regional analysis, followed by Top/Bottom state comparison, and concluding with quarterly and monthly consumption trends. This structured presentation helps users understand the complete analysis and derive meaningful insights from the electricity consumption dataset.

Story Components

- Dashboard 1 – State and Regional Electricity Consumption Analysis
- Dashboard 2 – Region-wise and Top/Bottom State Analysis
- Dashboard 3 – Quarter-wise and Month-wise Electricity Consumption Analysis

Story Screenshot

Story on electricity Consumption



Project Outcome

The project successfully demonstrates the use of Tableau Desktop for analyzing electricity consumption patterns through interactive dashboards and a Tableau Story. The developed solution enables users to compare electricity consumption across different states, regions, months, quarters, and years, identify high and low electricity-consuming states, and analyze the impact of the COVID-19 lockdown on electricity usage. The interactive dashboards and story provide meaningful insights that support informed decision-making for electricity boards, researchers, policymakers, and energy management organizations.

Tableau Public Publishing

After successfully developing the interactive dashboards and Tableau Story, the complete workbook was published on **Tableau Public**. Publishing the workbook enables users to access and interact with the visualizations online without requiring Tableau Desktop. The published workbook includes all three dashboards and the Tableau Story, allowing users to explore electricity consumption patterns through interactive filters and visualizations.

Tableau Public Link

The published Tableau workbook can be accessed using the following link:

https://public.tableau.com/app/profile/jagarlamudi.sree.harshini/viz/Electricity_Consumption_Analysis_17837570329990/StoryonelectricityConsumption

Tableau Public Share Link

https://public.tableau.com/shared/SNSZY5JMS?:display_count=n&:origin=viz_share_link